# Assignment-1

1. What is Python and Why is it Called a Interpreted language?

→ Python is high-level, General-Purposes programming Language used for web development, data science, automation, artificial Intelligence, and Many other applications.

→ It is Called an Interpreted language Because the Code is executed line-by line by an Interpreter Instead of being Converted into Machine Code before execution.

Ex:

* Python Code → Python Interpreter → Execution.

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2. What are the Key features of Python that make it Popular for Beginners and Professionals?

1. Simple Syntax – Easy to Read & Write
2. Interpreted language – No Need for Compilation.
3. Dynamically typed – No Need to declare Variable type.
4. Large Standard library – Many built-in Modules.
5. Cross-Platform – Works on windows, Linux, Mac
6. Object-Oriented – Supports OPP Concepts.

3. Difference between Python2 and Python3?

| Features | Python 2 | Python 3 |
|---|---|---|
| → Release Year | → 2000 | → 2008 |
| → Print Statement | → Print "Hello" | → Print ("Hello") |
| → UniCode Support | → Limited | → Full UniCode Support. |
| → Integer division | → Returns Integers | → Returns float |
| → Future Support | → No longer supported (ended 2020) | → Actively supported. |

4. What are Python's applications in Real-world Projects?

1. Data science
   * Data analysis, Machine learning, AI Models
   Libraries: Pandas, Numpy, Scikit-learn, Tensorflow.

2. Web Development:
   * Backend development
   * Framework: Django, Flask.

3. Automation:
   * Automation repetitive tasks
   * Web scraping

4. Data Analytics:
   * Data Cleaning, Dashboard Preparation.

5. Cyber Security:
   * Penetrating testing tools

5. What is PEP 8 and Why is it important in Python Programming?

→ PEP 8 Stands for Python Enhancement Proposal 8.

→ It is the offiCial Style guide for writing Python Code.

→ It defines Rules like:

- Indentation (4 Spaces)
- Variable Naming Conventions
- Maximum line length (79 Characters)
- Proper Code formatting

EX:

```
X = 10
Y = 20
Print (X + Y)
```

Make Python Code Clean, Readable, and Consistent.

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6. Who developed for Python and in Which Year was it Released?

→ Python Was Created by Guido Van Rossum

→ First Released: February 20, 1991

Developed at Centrum Wiskunde & Informatice (CW), Netherlands.

7. What does "dynamically typed" in Python?

→ In Python, you do not declare the data type of a variable.

→ The interpreter automatically determines the type during execution.

Ex:

```
x=10          # integer
x="Hello"     # now string
```

→ The same variable can store different data types.

8.) Difference between Compiler and Interpreter

| Compiler | Interpreter |
|---|---|
| → Converts entire program into Machine Code | → Executes code line by line |
| → Faster execution. | → Slower execution. |
| → Errors shown after compilation. | → Errors shown immediately |
| → Example: C, C++ | → Example: Python. |